

MACKAY MO, AustraliaWMO: **943670**Lat: **21.1172S**Lon: **149.2169E**Elev: **36**StdP: **100.89**Time Zone: **10.00 (E10)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	11.1	1.8	4.3	15.0	4.0	5.1	14.9	10.1	18.9	9.3	18.7	4.0	230	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.7	32.1	25.6	31.1	25.2	30.4	24.8	27.0	30.1	26.4	29.3	25.9	28.7	5.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	28.7	25.6	20.9	28.2	25.1	20.3	27.8	85.4	30.3	82.7	29.5	80.5	28.8	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.3	8.5	6.8	33.9	1.7	1.1	5.6	34.7	4.6	35.3	3.6	35.9	2.4	36.8		
(5)			4.7	28.1	1.5	1.0	3.6	28.8	2.7	29.4	1.9	30.0	0.8	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	26.9	26.7	25.7	23.6	20.7	18.3	17.4	18.4	21.1	23.7	25.3	26.7	
	DBStd	3.85	1.39	1.28	1.37	1.47	2.00	2.30	2.24	2.11	1.99	1.80	1.53	1.56	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	106	0	0	0	0	6	29	43	26	3	0	0	0	
	CDD10.0	4690	526	467	488	408	332	249	231	259	332	423	458	517	
	CDD18.3	1753	267	234	229	158	79	28	16	26	84	165	208	258	
	CDH23.3	13347	2684	2292	1953	833	179	26	15	64	323	909	1573	2497	
CDH26.7	2968	720	572	355	62	6	1	1	3	34	154	348	712		
Wind	WSAvg	4.7	4.9	5.0	5.2	5.4	4.9	4.7	4.3	4.1	4.1	4.5	4.6	4.7	
Precipitation	PrecAvg	1591	274	316	288	143	106	63	40	26	20	35	92	180	
	PrecMax	2847	1034	1071	921	521	546	392	269	122	234	152	563	1159	
	PrecMin	795	15	29	36	9	16	2	1	1	0	0	0	0	
	PrecStd	526	227	234	225	98	82	63	44	29	33	37	108	178	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	32.9	31.5	29.6	27.4	25.7	25.0	27.0	29.1	30.7	31.7	33.5	
		MCWB	26.3	27.1	26.3	24.6	22.3	21.3	20.2	20.9	22.0	23.7	24.0	25.6	
	2%	DB	31.7	31.1	30.2	28.2	26.0	23.9	23.2	25.1	27.7	29.5	30.6	32.1	
		MCWB	25.8	26.1	25.4	23.5	21.6	20.2	19.3	19.3	21.3	22.9	23.9	25.2	
	5%	DB	30.7	30.2	29.3	27.2	25.1	22.9	22.2	23.6	26.4	28.4	29.6	31.0	
		MCWB	25.4	25.4	24.7	23.0	21.1	19.8	18.8	18.7	20.7	22.4	23.3	24.8	
	10%	DB	29.8	29.5	28.5	26.5	24.2	22.1	21.3	22.5	25.1	27.3	28.7	30.0	
		MCWB	24.9	25.0	24.2	22.7	20.6	19.2	18.2	18.5	20.1	21.8	22.8	24.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	28.1	27.2	25.7	24.0	22.7	21.6	22.0	23.4	25.0	25.7	27.2
			MCDB	31.0	31.4	30.2	28.0	25.8	24.2	23.4	25.0	26.9	29.3	29.8	31.1
2%		WB	26.9	26.9	26.3	24.7	22.8	21.5	20.6	20.8	22.5	24.0	24.9	26.3	
		MCDB	30.0	29.8	28.9	26.7	24.9	22.9	22.1	23.2	25.6	27.4	28.7	29.9	
5%		WB	26.2	26.2	25.7	24.0	22.0	20.7	19.9	20.1	21.8	23.4	24.2	25.6	
		MCDB	29.2	28.9	28.1	26.2	24.2	22.2	21.2	22.4	24.8	26.8	28.0	29.2	
10%		WB	25.7	25.8	25.1	23.4	21.3	20.0	19.2	19.4	21.1	22.7	23.6	25.1	
		MCDB	28.6	28.3	27.5	25.7	23.5	21.5	20.8	21.6	24.0	26.1	27.3	28.6	
Mean Daily Temperature Range		5% DB	MDBR	5.7	5.4	5.4	5.7	6.9	7.1	7.9	8.6	8.4	7.4	6.8	6.6
			MCDBR	7.2	6.6	6.6	6.7	7.6	7.5	8.5	10.2	10.1	8.8	8.1	8.1
	MCWBR		3.2	3.2	3.2	3.5	4.3	4.7	5.4	5.8	4.9	3.8	3.3	3.2	
	5% WB	MCDDBR	6.3	6.1	5.9	5.7	6.6	6.2	6.9	8.2	8.1	7.7	7.3	7.3	
		MCWBR	3.1	3.3	3.2	3.2	3.9	4.2	4.7	4.9	4.1	3.6	3.3	3.3	
Clear-Sky Solar Irradiance	taub	0.403	0.396	0.384	0.361	0.342	0.331	0.316	0.324	0.353	0.382	0.393	0.397		
	taud	2.517	2.540	2.557	2.606	2.614	2.637	2.661	2.625	2.531	2.476	2.482	2.503		
	Ebn at Noon	942	937	923	906	886	879	906	930	938	939	946	948		
	Edh at Noon	114	109	103	92	84	79	79	88	104	115	117	115		
All-Sky Solar Radiation	RadAvg	6.34	5.96	5.50	4.87	4.20	3.72	4.21	5.06	6.14	6.77	6.91	6.88		
	RadStd	0.47	0.65	0.52	0.47	0.27	0.37	0.32	0.35	0.42	0.39	0.80	0.73		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page